UNIVERSITY OF CENTRAL FLORIDA

EEL6812 Neural Networks and Deep Learning

Assignment 3: Recurrent Neural Networks, Transformers, and Variational Autoencoders

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EEL6812 Neural Networks and Deep Learning

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Contents

Contents **1**

1 Sequence Classification Using Recurrence Models **2**

1.1 Recurrent Neural Network Performance 2

2 Character Generation Using Transformer Model **5**

2.1 Dimensionality 5

2.2 Positional Encoding 5

2.3 Multi-Headed Attention 10

2.4 Number of Layers 16

2.5 Context Length 20

2.6 Causal Masking 24

2.7 Model Summary 26

3 Generative Modeling Using Variational Autoencoders **28**

3.1 VAE Training 28

3.2 Generate New Samples 29

3.3 Modify the Loss Function 30

3.4 Latent Space Dimension 31

1

Sequence Classification Using Recurrence Models

In this task, I implemented a recurrent sequence classifier and analyzed the performance of three different RNN models. To avoid processing padded tokens, I used the `pack_padded_sequence` and `pad_packed_sequence` to pack the input and unpack the output, respectively.

1.1 Recurrent Neural Network Performance

The LSTM, RNN, and GRU networks were trained for 50 epochs using a batch size of 32 and a learning rate of 10^{-3}. A summary of the final training metrics can be found in Table 1.1. Figures 1.1- 1.4 show the training loss, training accuracy, testing loss, and testing accuracy across epochs for each of the three networks. The performance and accuracy of LSTM and GRU models is comparable, with the GRU reaching the highest testing accuracy of 99.90% on epoch 43. Meanwhile, the RNN is nearly 10% less accurate than the LSTM and GRU models for sequence classification. The highest testing accuracy reached by the RNN model is 91.60%, which in Figure 1.4 almost immediately drops to an accuracy below 60% at epoch 30. This is most likely due to the RNN's inability to process long-term relationships in the Reber sequences.

Table 1.2 summarizes the number of learnable parameters in each network. The LSTM and GRU's larger parameter count helps to explain their ability to better classify the Reber sequences than the RNN. These parameters are used in their gating mechanisms, which help preserve and update longer-range information more effectively than a standard RNN.

Network	Training loss	Training accuracy	Testing loss	Testing accuracy
LSTM	0.0058 (48)	99.92% (45)	0.0049 (47)	99.85% (47)
RNN	0.2728 (26)	90.45% (26)	0.2416 (26)	91.60% (26)
GRU	0.0046 (47)	99.93% (46)	0.0044 (46)	99.90% (43)

Table 1.1: *Summary of final training metrics for the three recurrent neural network architectures.*

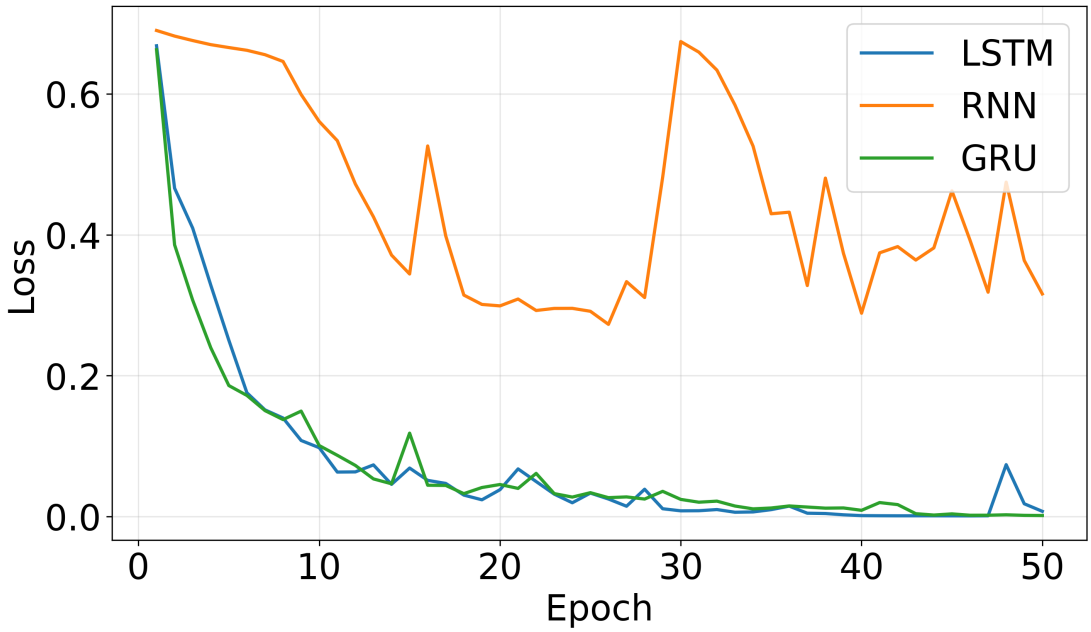


Figure 1.1: *Training loss for LSTM, RNN, and GRU*

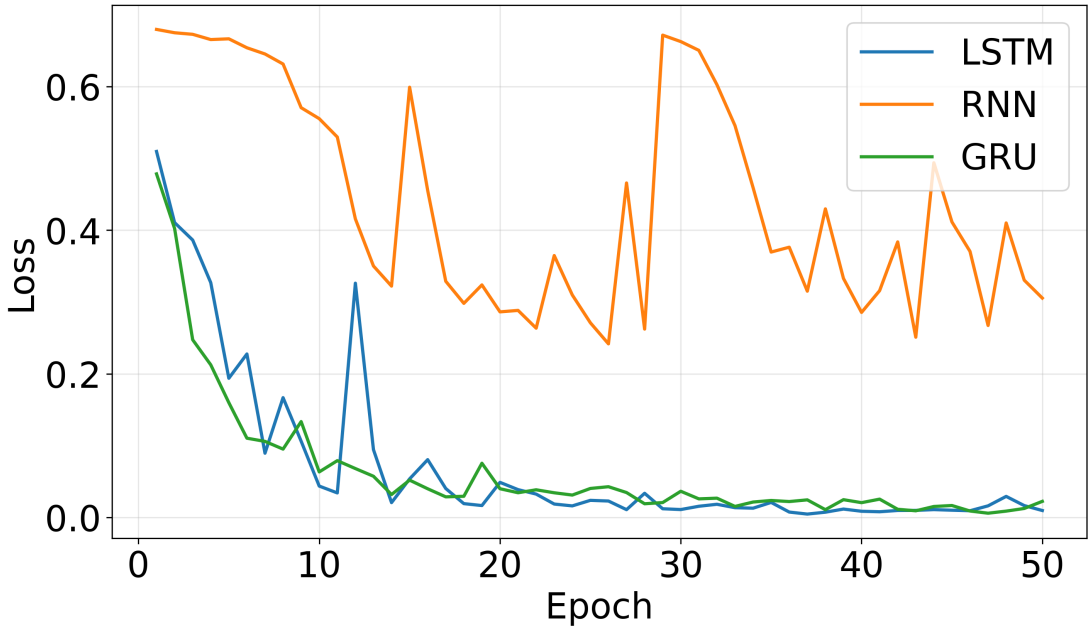


Figure 1.2: *Test loss for LSTM, RNN, and GRU*

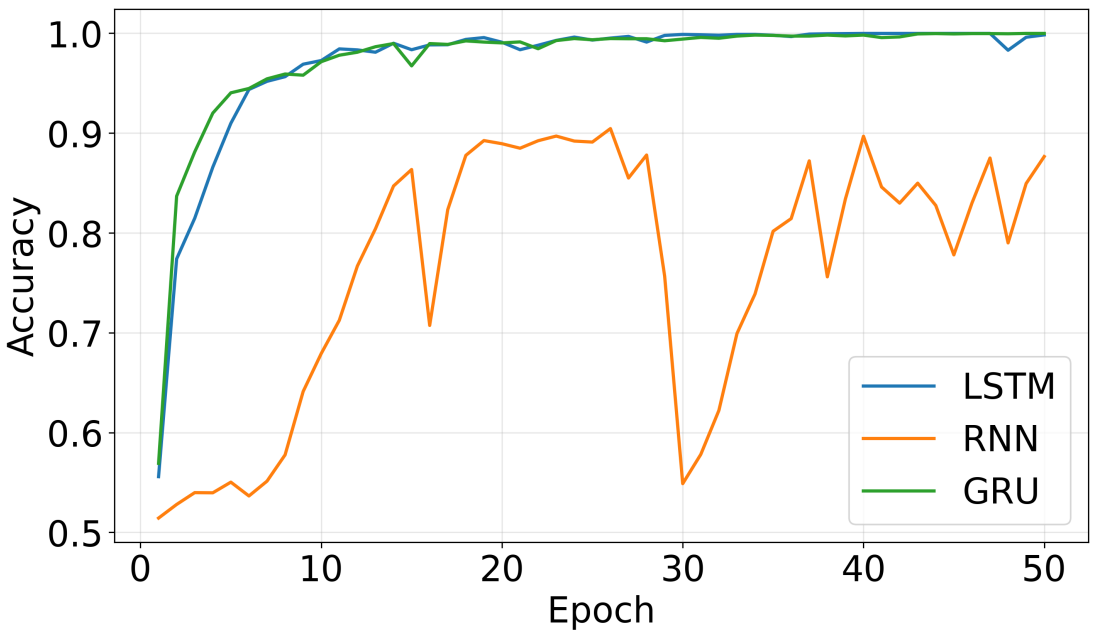


Figure 1.3: *Training accuracy for LSTM, RNN, and GRU*

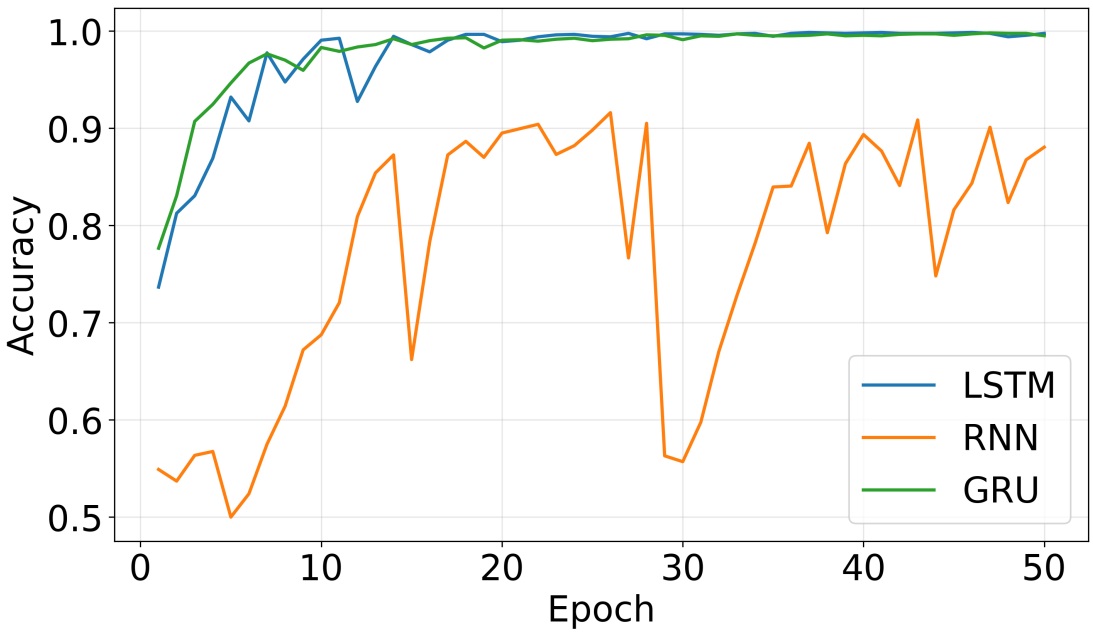


Figure 1.4: *Test accuracy for LSTM, RNN, and GRU*

Network Type	Embedding Layer Dim	Network Dim	Linear Layer Dim
LSTM	(8,32) with 256 params	(32,64) with 25,088 params	(64,2) with 130 params
RNN	(8,32) with 256 params	(32,64) with 6,272 params	(64,2) with 130 params
GRU	(8,32) with 256 params	(32,64) with 18,816 params	(64,2) with 130 params

Table 1.2: *Layer dimensions and parameter counts for recurrent networks*

2

Character Generation Using Transformer Model

In this task, we train a character-level transformer model to generate text data. The following sections detail some ablation studies using the transformer model.

2.1 Dimensionality

A single dataset sample consists of two 1D tensors, x and y, each of length block size, which is the context length. For a training batch, this is the 2D tensor [batch size, block size]. The default settings I'm using for training are a batch size of 64 and a block size of 64, so the batched training tensor is of size [64,64]. Since the goal of the transformer network is to predict the next character in the sequence, the label y should necessarily be shifted one character to the right.

2.2 Positional Encoding

Tokens receive position encodings to denote the structural order of a sequence, as tokens are not processed one at a time like in the RNN. Positional encodings are applied to the entire input sequence. The dimension of one positional encoding vector is [block size, 128], and the dimension of the full positional encoding tensor is [batch size, block size, 128], where the default values are a batch size and block size of 64.

After training the models for 10 epochs, the final validation loss was ~1.4 with positional encodings and ~1.73 without positional encodings. In general, the validation loss across epochs was a perfectly offset curve, with the model containing no positional embeddings always having a higher loss.

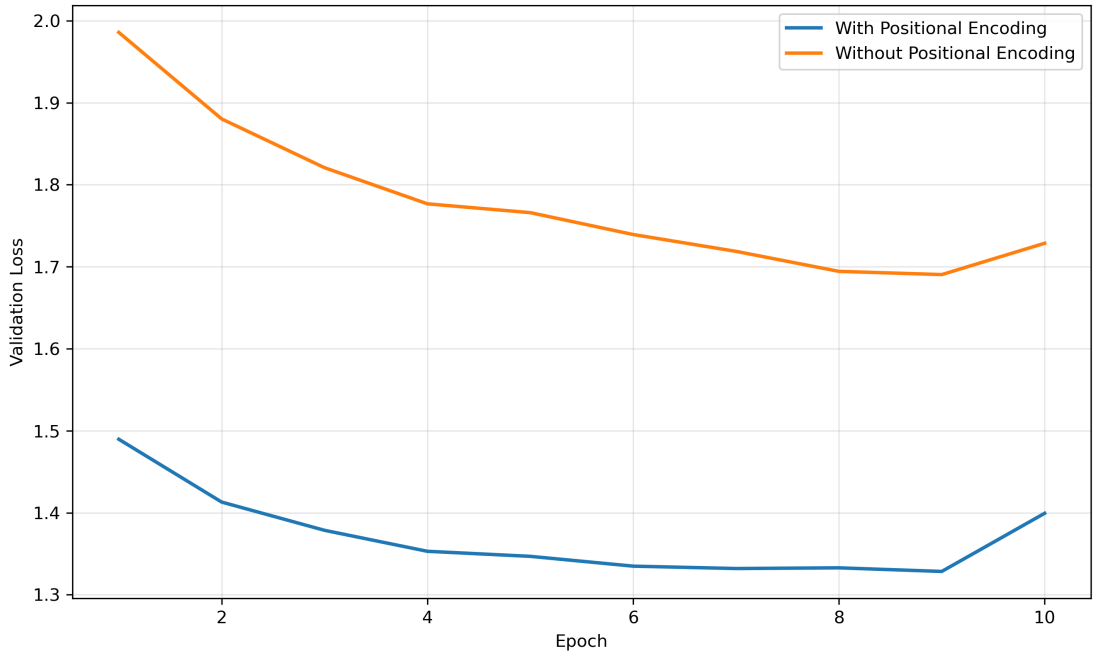
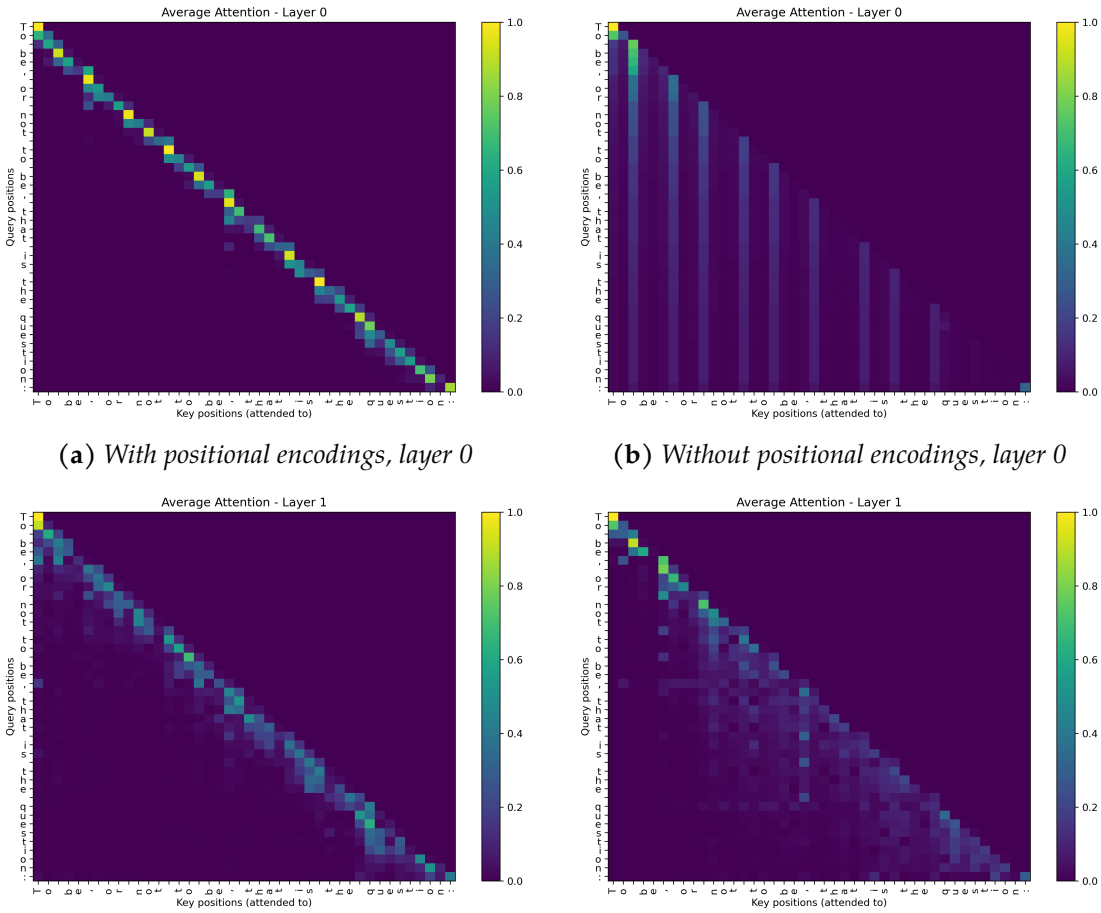


Figure 2.1: *Validation loss curves with and without positional encodings*

From the average attention maps for layers 0 and 1, we see the necessity of positional encodings. In the first layer, the attention map with positional encodings have a fairly linear correlation between query positions and key positions, and many keys are strongly attended to. For the first layer without, there is no linear correlation or strong attentiveness. Query positions have virtually no long-range attentiveness and attend mostly to characters nearby, organized into columns. Even on layer 1, the strongest attention is seen at the beginning of the sequence, with keys that are nearby in position.

Qualitatively, the model outputs without positional encodings are gibberish: some names are spelled correctly, and there is some hint of sentence structure, but everything else is completely incomprehensible. Meanwhile, the outputs with positional encodings are quite good, with most of the words being spelled properly or discernable.

```
Prompt: "ROMEO:"
Output: "Our news in seath, as we abend, quit you many no off!
What, God, even, say 'twas me to expect;
Be all our against his fame weous none; but sees of where wonder with
    thy parts:
This? Bell'st whench let, avake one
thee queen thou art-givert.
```



(**a**) *With positional encodings, layer 0*

(**b**) *Without positional encodings, layer 0*

(**c**) *With positional encodings, layer 1*

(**d**) *Without positional encodings, layer 1*

Figure 2.2: *Average attention maps for layers 0 and 1 of transformer model trained with and without positional encodings.*

```

COMINIUS:
But tell him by his eage.

AUTOLYCUS:
Come, thou"
```

Listing 2.1: *Prompt 1 for default model parameters.*

```
Prompt: "JULIET:"
Output: "And unwill we died unwith you;
There mause, the portent of your uncle.

CAPULET:
Good.

CORIOLANUS:
My lord, son child will I will.

ISABELLA:
Me such stand, my lord, let's news the pardon'd your father of;
And bad with heir follow. Go may death, do medness,
Smalital, I think by their teach-blament?"
```

Listing 2.2: *Prompt 2 for default model parameters.*

```
Prompt: "To thine own self be true, And it must follow, as the night
    the day, Thou canst not then be false to any man."
Output: "DUKE VINCENTIO:
Sir, and my glow together?
When I am heard, no more pain again. The Duke of York;
Thyself is the strike gracious blunt death,
Than I was a wrong warch'd know'st not life;
And 'tis comeful unto behave it my heart,
It live to her. There he had in this sunder:
O, Richard had 'scat beat"
```

Listing 2.3: *Prompt 3 for default model parameters.*

```
Prompt: "ROMEO:"
```

```
Output: "Make my brother; shall life, shall I ame grace
ten seet thme wigh pecears an that mat adm the,
Thule ndedroucld him? he ussut-

WARWICK:
And hat it bod, nesar hand coves ar and chis laitte,
Of Bellengants ord n
Thivet theeep tikes t hesourst oit
Biong trropanth, mmal ltraneaw spison."
```

Listing 2.4: *Prompt 1 for default model parameters without positional encoding.*

```
Prompt: "JULIET:"
Output: "I an h
JULIET:
Inater am fit a his.

VOLUMNIA:
I hat to, f hey, lover you beeaste piale comes of e let
To-bre madedspl. oved viner'd trut
tour Wistheam; chiffofor wilse. Sis The Brebetheing their
Thow, Richbard ledron, ip. hisous which miselvikeng Rome by theyou ha
For they wou hay.

ANGELO:
And no whilsh, iply yo"
```

Listing 2.5: *Prompt 2 for default model parameters without positional encoding.*

```
Prompt: "To thine own self be true, And it must follow, as the night
    the day, Thou canst not then be false to any man."
Output: "Good preciman King Ed; whom I have duke have sol the young
    the kiglf:
Caris; browsay ry ste benutes Marcious y sone recown bling the fanure
     anw:
I'll thaver fom miser but than bege
veath seend othath: bus w, hich ane with our othat;
Dist o my like.
Haing again! s Anow THe you spere it
```

```
Whatcho er wo"
```

Listing 2.6: *Prompt 3 for default model parameters without positional encoding*

2.3 Multi-Headed Attention

In this section, the number of attention heads in each layer is swept for values of [1,2,4,8]. The average entropy over all query positions, that is, the average of the entropies of each row, can be computed by taking the average of

$$H_i = -\sum_j A_{ij} \log A_{ij}$$

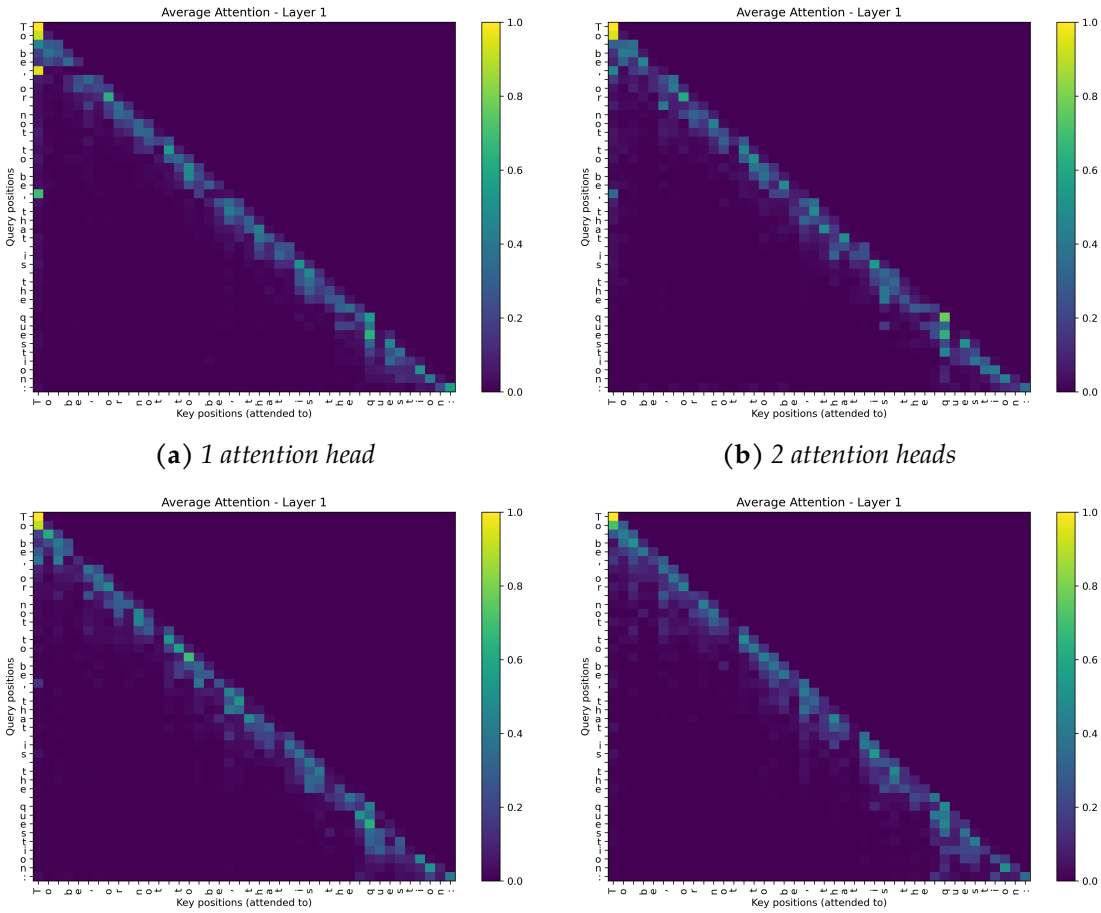
The full average entropies for each training configuration can be seen at the end of this section in Table 2.1. In general, as the number of attention heads increases, the average entropy per layer decreases, with the 1-headed model having an average entropy of 0.9067 in layer 0 and the 8-headed model having an average entropy of 0.3837 in layer 0. The reason for this decrease is that additional attention heads can allocate attention more selectively. Visually inspecting the attention maps in Figure 2.3, query positions below the diagonal increasingly attend to key positions as the number of attention heads is increased.

Qualitatively, the generated outputs increase in quality as the number of attention heads increases. For a single attention head, the outputs are borderline incomprehensible, especially Prompts 2 and 3. By 8 attention heads, the responses are much more interpretable.

```
Prompt: "ROMEO:"
Output: "The Warwick and Duke of Sun.
Do may besecur.

Second Servant:
Discosss! the paintinue, lend my my son.

BRUTUS:
Weor ded trobber
Indeed him with that are it
The more, we'll perfeets
```



(**a**) *1 attention head*

(**b**) *2 attention heads*

(**c**) *4 attention heads*

(**d**) *8 attention heads*

Figure 2.3: *Average attention map for layer 1 of model trained with various attention heads.*

```
For means to a purn fault you.
EDWARD:
Theres very our life, I little of confix me be
Hysecuted in screals prove mend"
```

Listing 2.7: *Prompt 1 for default model parameters with 1 attention head.*

```
Prompt: "JULIET:"
Output: "Ay, we they gracious like passed, I will a laster'd.

GLOUCESTER:
You crue as trabble, lord, I we me for Pomfretiges young:
Tand your bowers tander unjugnances, I ink your gallange, or with us,
I we case of this men. But was 'twere the dead.

QUEEN MARGARET:
Yes, regars, none since? I gripeness
Came"
```

Listing 2.8: *Prompt 2 for default model parameters with 1 attention head.*

```
Prompt: "To thine own self be true, And it must follow, as the night
    the day, Thou canst not then be false to any man."
Output: "And detender to enough gods; no
tis for mine our friarl it, find me cape,
By me to be more than to mean strange he,
When irove thee, peaces my town were not condamn's.

KING RICHARD III:
Forsphere's death, he is her heart; we thou haspeed by attent
Like the heaven, a't me from stands in coomefit:
My"
```

Listing 2.9: *Prompt 3 for default model parameters with 1 attention head.*

```
Prompt: "ROMEO:"
Output: "Curst, what he nows will tents,
In any the carpear by gone, my lord.

QUEEN MARGARET:
```

```
See use the rottlen you with you, I reaten
By time to queen, oo'er news.
To be the next his throse shall
To svearity in king against see his lawful be,
Youngs queen, his stay nature man fen,
as inforcet you, father"
```

Listing 2.10: *Prompt 1 for default model parameters with 2 attention heads.*

```
Prompt: "JULIET:"
Output: "Was you must thereby, I think, unbance,
That I will enemies hath none since debtly with dost again.

SICINIUS:
I have a king pale his crue? Edward it loved,
Wonder well deliver! Clarence for duke
And I turn besies 'Consent good Captsatio, and
life, you was meeth Edward: 's yet not
say this rock, and"
```

Listing 2.11: *Prompt 2 for default model parameters with 2 attention heads.*

```
Prompt: "To thine own self be true, And it must follow, as the night
    the day, Thou canst not then be false to any man."
Output: "What would love these would would pay,
How may poll then then sons; but thought it think us land forget to
    them?

Nurse:
Sprinces, that wound, and the king ship to proloud his
tock. If respected. For Romeo, my lord,
Beseech your graces, and it but our promise hence,
Our bagot other, the straight hat"
```

Listing 2.12: *Prompt 3 for default model parameters with 2 attention heads.*

```
Prompt: "ROMEO:"
Output: "Our news in seath, as we abend, quit you many no off!
What, God, even, say 'twas me to expect;
Be all our against his fame weous none; but sees of where wonder with
     thy parts:
```

```
This? Bell'st whench let, avake one
thee queen thou art-givert.

COMINIUS:
But tell him by his eage.

AUTOLYCUS:
Come, thou"
```

Listing 2.13: *Prompt 1 for default model parameters with 4 attention heads.*

```
Prompt: "JULIET:"
Output: "And unwill we died unwith you;
There mause, the portent of your uncle.

CAPULET:
Good.

CORIOLANUS:
My lord, son child will I will.

ISABELLA:
Me such stand, my lord, let's news the pardon'd your father of;
And bad with heir follow. Go may death, do medness,
Smalital, I think by their teach-blament?"
```

Listing 2.14: *Prompt 2 for default model parameters with 4 attention heads.*

```
Prompt: "To thine own self be true, And it must follow, as the night
    the day, Thou canst not then be false to any man."
Output: "DUKE VINCENTIO:
Sir, and my glow together?
When I am heard, no more pain again. The Duke of York;
Thyself is the strike gracious blunt death,
Than I was a wrong warch'd know'st not life;
And 'tis comeful unto behave it my heart,
It live to her. There he had in this sunder:
O, Richard had 'scat beat"
```

Listing 2.15: *Prompt 3 for default model parameters with 4 attention heads.*

```
Prompt: "ROMEO:"
Output: "Here and feall bear;
And, see answer have love it so, thy heart his crown;
Was says a more where sturn haft to Clifford of stir;
And with about hear the grace coldering pity
entence to death: already, or this said so That Duke o' the
    scopperform maid.

CORIOLANUS:
I
My fancy undred the banished eagl"
```

Listing 2.16: *Prompt 1 for default model parameters with 8 attention heads.*

```
Prompt: "JULIET:"
Output: "She would keep to propose me stept, blood:
It is a man; him asleep. Let him: reason of
you are was; but to part.

DUKE VINCENTIO:
And are not me, by the palmer; you hadst been you.

HERMIONE:
Not I tear; affect me, I am good love?  here is the friendly-?
I am foreter than horse, therefore me,  art o"
```

Listing 2.17: *Prompt 2 for default model parameters with 8 attention heads.*

```
Prompt: "To thine own self be true, And it must follow, as the night
    the day, Thou canst not then be false to any man."
Output: "Not royal foe will dear?

Post:
Ah, vice, dispance, when cannot the time of
earth of York; away, justice.

PERDITA:
My lord, sir!
Why, waw.
```

```

LEONTES:
You be shall he faithouse's most kill.
O, but I say the dream--this deposition
A standers of she's agrees, but your out, your come and pleasure.

CLAR"
```

Listing 2.18: *Prompt 3 for default model parameters with 8 attention heads.*

2.4 Number of Layers

In this study, I test three values for the number of attention layers: [1,2,4]. The trend seems to be that deeper models produce lower validation loss than shallower models, as seen in Figure 2.4.

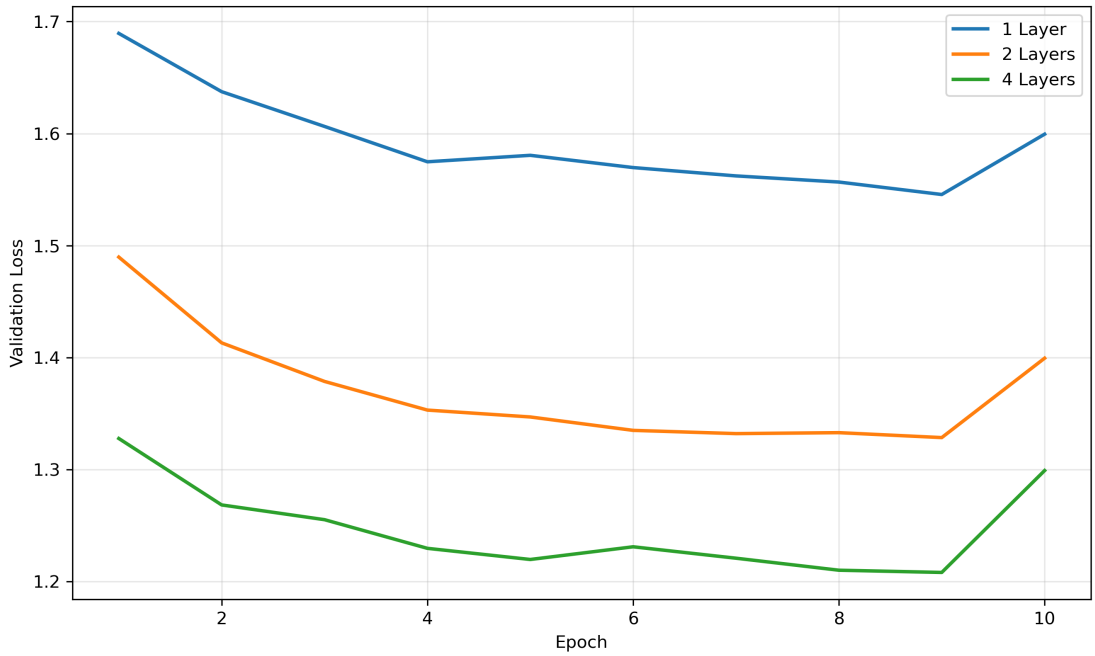
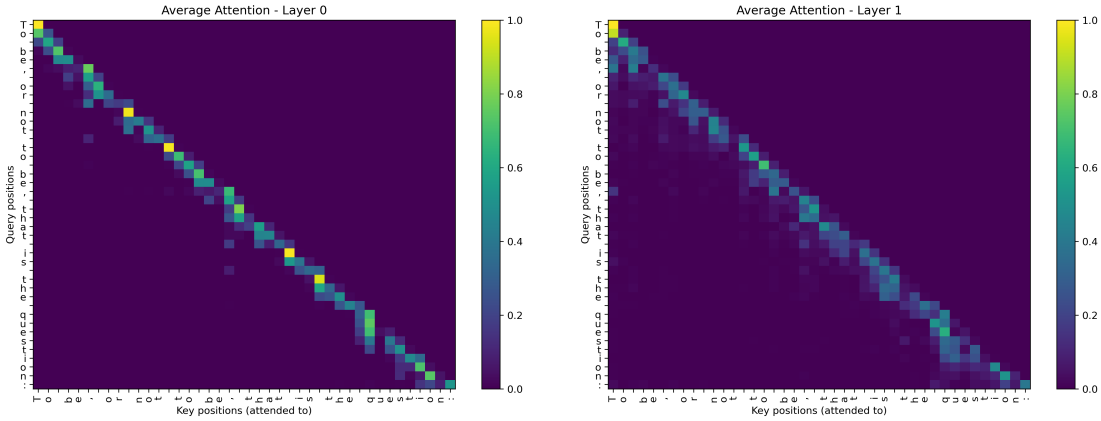
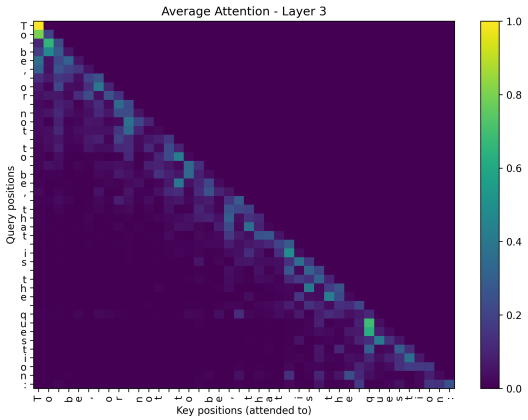


Figure 2.4: *Validation loss across epochs for models with 1, 2, and 4 attention layers.*

From the attention maps in Figure 2.5, query positions tend to attend to more key positions as the depth increases. However, the intensity (closeness to 1) of the attentiveness is lower on average for deeper layers. This suggests that deeper layers draw more context from a wider range of positions, not just attending to neighboring tokens as in a single attention layer.



(**a**) *1 attention layer* (**b**) *2 attention layers*



(**c**) *4 attention layers*

Figure 2.5: *Average attention maps for the first head of the final attention layers for models with 1, 2, and 4 attention layers.*

While the generated outputs are grammatically coherent with a single layer, the sentences make little sense in context. Prompt 3 highlights this especially. For deeper models, while the coherence isn't perfect, it definitely improves, and the quality of spelling becomes much better.

```
Prompt: "ROMEO:"
Output: "Lay
The posse on thank well, I strong that place,
The you stoose were rement
To make as upon bears the majesty, lawn.

LUCIO:
We wrap I than her soceet, I'll be suness
To told I honoury house queen:
And in they will a trance's pophen the bravour pamp'd subs you I dost
```

```
    your did her.

ISABELLLA:
Ear g"
```

Listing 2.19: *Prompt 1 for default model parameters with 1 attention layer.*

```
Prompt: "JULIET:"
Output: "I do whose to your fair their virtuer is might
    happentingsoulow were farther
Than very . Go mive meand
togethou dilly your a thou, should be ducheek:
And it here?
And day will to sey. He have to the time neither of not his of?

HORTHUMBERLAND:
Why, to I; he promport-Fon I can: ilain, what I been I g"
```

Listing 2.20: *Prompt 2 for default model parameters with 1 attention layer.*

```
Prompt: "To thine own self be true, And it must follow, as the night
    the day, Thou canst not then be false to any man."
Output: "
ANGELO:
Only
A from to Can Frue all prince, to lamus the word,
This boymes thee time to first to the unour:
Thome my credink; en present cill sove that find of thee a dard 'gain
    who love bunds.

MENENIUS:
Well, ho.
Hop not 'em, find now commod else, I
GLOUCESTER:
You serving to bring: I ushould cam"
```

Listing 2.21: *Prompt 3 for default model parameters with 1 attention layer.*

```
Prompt: "ROMEO:"
Output: "And with the prince cave no more Polixenes
```

```
Hath rush'd me then sweet sound wherefore.

Grace Paulina cousin?

ESCALUS:
One end, as I like a way or taken an old
trust, and your entreat father, wood, hark!
Come refign.

CORIOLANUS:
Proclaim Xgham,
My behalf I cannot too lips.'

BUCKINGHAM:
O, I'll thr"
```

Listing 2.22: *Prompt 1 for default model parameters with 4 attention layers.*

```
Prompt: "JULIET:"
Output: "Madam, that would not, I do stay to thy business;
For it is the bave we ere all arms a prisoner
Hath the king in the heaven in a vermain.

KING RICHARD III:
Let them without the heavy cause to France;
I'll know you any one our harms in very
birding thus thou quiet be threefore:
Thou might spend a va"
```

Listing 2.23: *Prompt 2 for default model parameters with 4 attention layers.*

```
Prompt: "To thine own self be true, And it must follow, as the night
    the day, Thou canst not then be false to any man."
Output: "
DUKE VINCENTIO:
More I pity you, plaze;
It is a letter life, which he shall be determined.

KING HENRY VI:
```

```
Yes, Barnardine? Provost: dear-peace.

GLOUCESTER:

KING RICHARD III:
And where you?

HENRY BOLINGBROKE:
Go, till thou dost not accover: I have done we see,
A steel Lewis holy words, and made"
```

Listing 2.24: *Prompt 3 for default model parameters with 4 attention layers.*

2.5 Context Length

This section explores the impact of context length on model performance, using block sizes of [32,64,128]. From Figure 2.6, increasing the block size has the effect of reducing the validation loss across epochs. The impact of context length is generally constant across epochs for each loss curve.

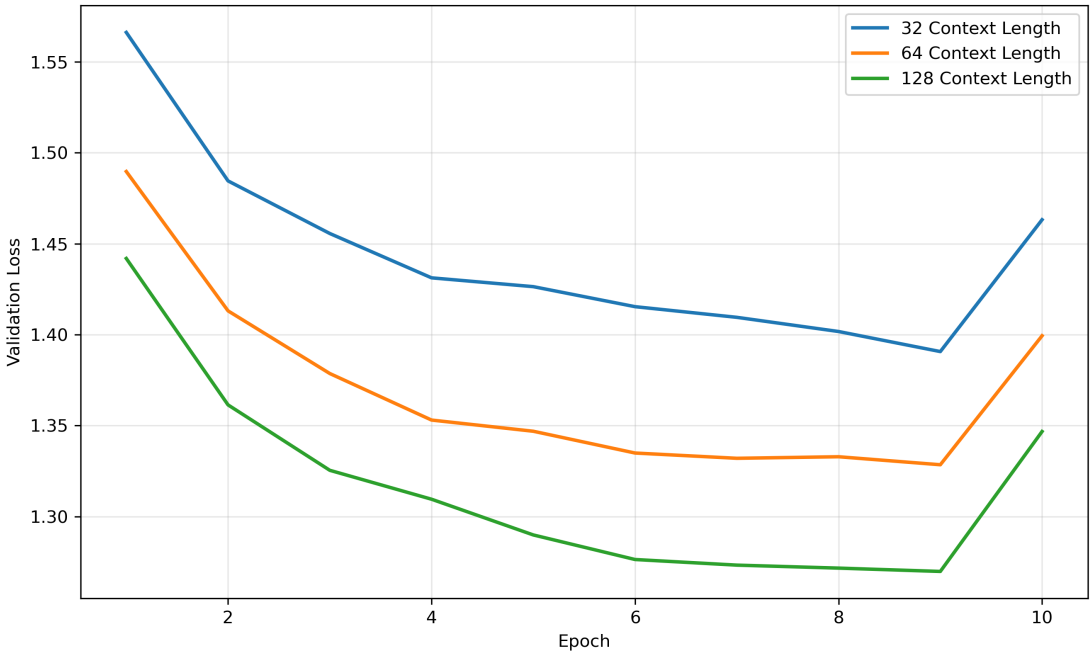
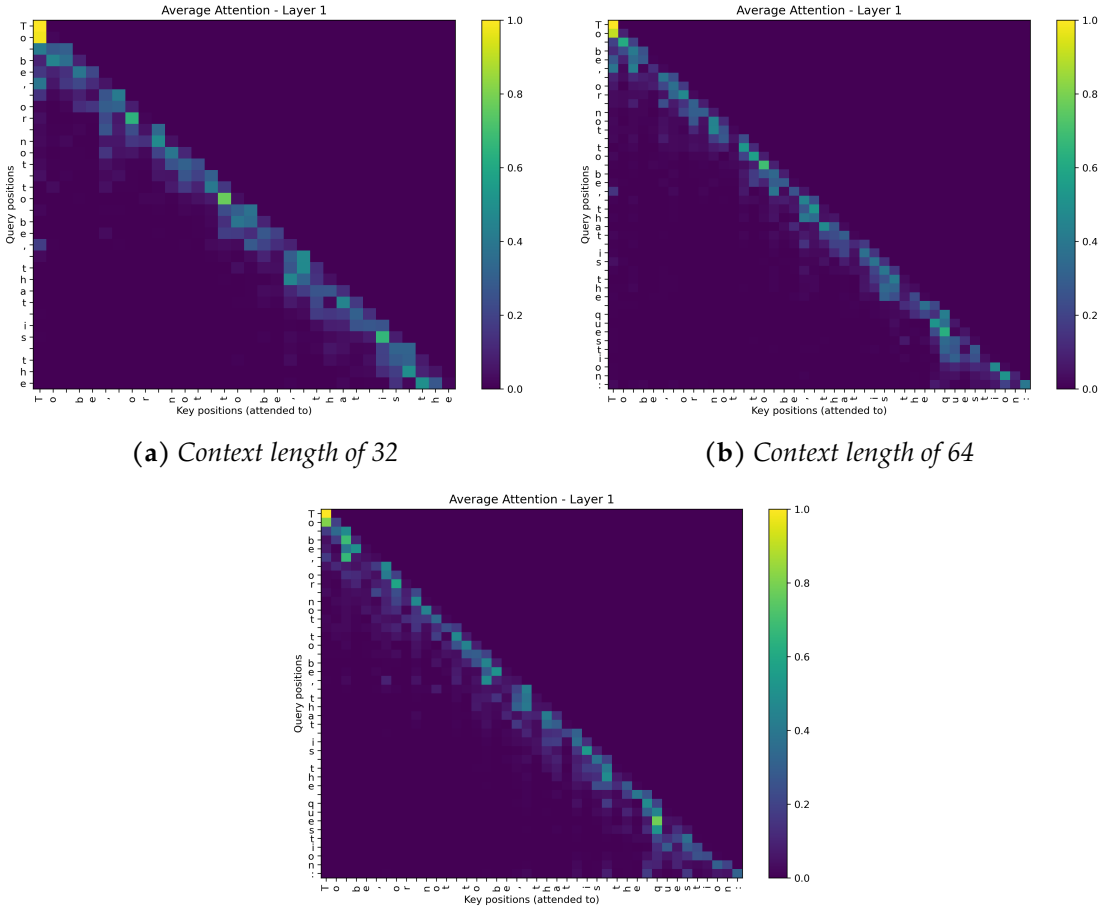


Figure 2.6: *Validation loss across epochs for context lengths of 32, 64, and 128.*

From the attention maps, an increased context length tends to increase the number of key positions attended to by a query position. However, it does not decrease the

attention score for the positions lying closest to the diagonal. Overall, increasing the context length causes more key positions to be attended to by more query positions.



(**a**) *Context length of 32*

(**b**) *Context length of 64*

(**c**) *Context Length of 128*

Figure 2.7: *Average attention maps for the second attention layer for context lengths of 32, 64, and 128.*

The generated outputs for all three context lengths generally have good grammar and interpretable words. The main impact of increased context length, it seems, is that the generated outputs more closely match the prompts. For example, with a context length of 32, prompting "Romeo" leads to an output that does not involve Romeo, and the same with Juliet. Meanwhile, with a context length of 128, the final prompt, which is a longer quote from one of Shakespeare's other works, leads to a output that speaks of nobility and honor. This output is more in context with the prompt than smaller context lengths were.

```
Prompt: "ROMEO:"
Output: "I could concling; most me from it.

```

```
MENENIUS:
Wilt not speak him stand to furiond Somerset,
And true so fair a truly, by heaving abroke.

LEONTES:
She with nere it, and they will not to be spark?

CLARENCE:
Thou would you tow joy!

QUEEN MARGARET:
I have to the rend!

AUTOLYCUS:
Your sing-like, an t"
```

Listing 2.25: *Prompt 1 for default model parameters with a context length of 32.*

```
Prompt: "JULIET:"
Output: "I'll tell it arm; and therefore what cannot so, shall not,
So forsward we have the princely,
Why, news is in Somerset, slaughter
To proud is day to say and lord, friar, be the rebel:
O three, sorrow? bark you go here?
I will proticulary the is it banishment,
Simp were disleep, have was it.

FRIAR LA"
```

Listing 2.26: *Prompt 2 for default model parameters with a context length of 32.*

```
Prompt: "To thine own self be true, And it must follow, as the night
    the day, Thou canst not then be false to any man."
Output: "I do voany datch'd theirl be never and
Op accomfort my body toat those more
Than here; to that you partings she peace is envy to suver'd; he
    fiery Belivex'd?
ell, pains, us in good, they is for the
such chyperity more boots.
```

```
Here is not grace, thy all body: he stard Gentlemen,
And send "
```

Listing 2.27: *Prompt 3 for default model parameters with a context length of 32.*

```
Prompt: "ROMEO:"
Output: "Read's ip it, painted.

MERCUTIO:
It shall we have been he need noble serve,
I will put sin wed by him mine own to thum.

LUCIO:
'Tis fall, and tender me no pardon, our sower;
Shepherd, Pompey, I
must convey'd ongly at to ken a have speech enious craffs?:
And, let me the which soils my boy abody.
he"
```

Listing 2.28: *Prompt 1 for default model parameters with a context length of 128.*

```
Prompt: "JULIET:"
Output: "Then, would not as excure this either breaster
Of throughly crown, a put of sir.

Nurse:
I have must been embroke as the city,
Friends is both.

LEONTES:
True being the salt within in this rine!
Be as the badies wind all not die incity.

POMPEY:
I'll you not be julity.

MERCUTIO:
The fliends for dau"
```

Listing 2.29: *Prompt 2 for default model parameters with a context length of 128.*

```
Prompt: "To thine own self be true, And it must follow, as the night
    the day, Thou canst not then be false to any man."
Output: "
ROMEO:
'Tis done:' any leized vex'd many house to
O night blood have the king noble Henry's more
A honour is cruel nation?
Fare, knows is't destroy, his prudes sink
For the since is of that and stand in sift.

BISHOP OF, Romeo, show not any joins glong:
Upon my heart done, cony Edward us own Romeo,
```

Listing 2.30: *Prompt 3 for default model parameters with a context length of 128.*

2.6 Causal Masking

Causal masking is used to prevent future information from leaking out during training. Each token should only attend to itself and earlier tokens, not future ones. If the model "looks ahead" when predicting the next character/word, the task is trivial. This is perfectly demonstrated by the average attention matrix for the second layer (layer 1) in Figure 2.8. Each character in the query rows attends most heavily and almost exclusively to the very next character in the key column, with an attention score close to 1.

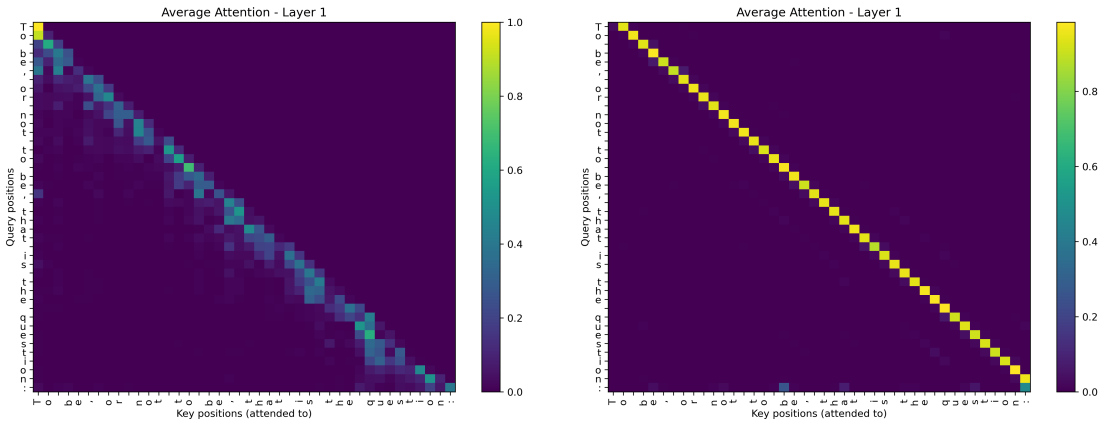


Figure 2.8: *Average attention maps for the second attention layer with (left) and without (right) causual masking.*

From a plot of the training and validation loss in Figure 2.9, the performance of the model with no causal masking seems significantly better than with, almost a loss of zero right away. However, inspecting the generated outputs shows that without causal masking, the generated text is almost incoherent. Sometimes, actual words are formed, but generally the sentences and outputs do not make sense. This makes sense if the query positions are only attending to their immediate neighbors, as seen in the attention map.

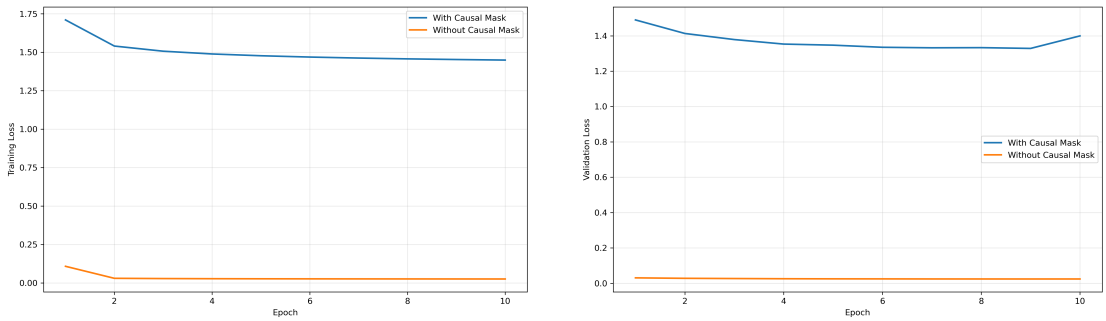


Figure 2.9: *Training loss (left) and validation loss (right) for masked and unmasked models.*

```
Prompt: "ROMEO:"
Output: 56 blank lines, then "KING HENRY VI:
He's is of Volice.

MICINIUS:
My die, that it to me, and them what all.

OXFORD:
What? not leble-morrow ach inclivet
The waken your bater or of heaves: whath teal;
Or bleavise, if news.

KING HENRY VI:
Romess! Ebour for a mother"
```

Listing 2.31: *Prompt 1 for default model parameters without causal masking.*

```
Prompt: "JULIET:"
Output: 56 blank lines, then "AST:
Mown the dready,
Which purpose to can of: gord!

```

```
LUCIO:
A honour mostrust. Give, unter not.

BRUTUS:
Nor sent upon and words bestlyf you both in all:
Myself shall meadly, cheer the dofford,
Dystal agoens that my brother comes:
My and there"
```

Listing 2.32: *Prompt 2 for default model parameters without causal masking.*

```
Prompt: "To thine own self be true, And it must follow, as the night
    the day, Thou canst not then be false to any man."
Output: "And distisday bless them of to than as Lace Angelo;
Their and post they fool, the rost him.
Why hath that queen in to protch
Unto your ove must your come. My my doubt in heart.
To Ladon then to be still his humble the threer and;
Consce, thee, your not comfort me;
For no bilixt aunt, cause thwy, all"
```

Listing 2.33: *Prompt 3 for default model parameters without causal masking.*

2.7 Model Summary

Finally, I present the validation loss and average entropy of each attention layer for each of the training runs conducted in this section in Table 2.1.

Parameters	Val. Loss	Layer 0 Entr.	Layer 1 Entr.	Layer 2 Entr.	Layer 3 Entr.
4, 2, 64, yes, yes	1.3994	0.4543	1.05264	-	-
4, 2, 64, no, yes	1.7286	2.2838	1.5850	-	-
1, 2, 64, yes, yes	1.4467	0.9067	1.5824	-	-
2, 2, 64, yes, yes	1.4130	0.5876	1.3781	-	-
8, 2, 64, yes, yes	1.3669	0.3837	0.9149	-	-
4, 1, 64, yes, yes	1.5994	0.4333	-	-	-
4, 4, 64, yes, yes	1.2990	0.4582	0.7529	1.1983	1.3080
4, 2, 32, yes, yes	1.4632	0.2575	0.8958	-	-
4, 2, 128, yes, yes	1.3467	0.6115	1.1274	-	-
4, 2, 64, yes, no	0.0242	1.9086	0.3141	-	-

Table 2.1: *Summary of final validation loss and average entropy per layer for all ablation studies. Parameter order: number of attention heads, number of layers, context length, positional encoding* (*yes*/*no*)*, and causal mask* (*yes*/*no*)

3

Generative Modeling Using Variational Autoencoders

3.1 VAE Training

The VAE model is first trained using a latent dimensionality of 20, 10 epochs, $\beta = 3$ for Beta-VAE, BCE loss, and KL divergence loss defined as

$$L = -\frac{1}{2}\sum_{j=0}^{k-1}(1 + \log(\sigma_j^2) - \mu_j^2 - \sigma_j^2)$$

KL divergence loss starts out small and increases slightly over epochs until saturating. KLD loss serves as a regularization term, which penalizes the latent distribution from drifting too far from a chosen prior, which is the normal distribution in a VAE. β-VAE gives the regularization term an adjustable weight, which is typically > 1 to keep the latent distribution closer to the prior. In general, the loss value saturated around epoch 10, with only margin decreases in loss in future epochs.

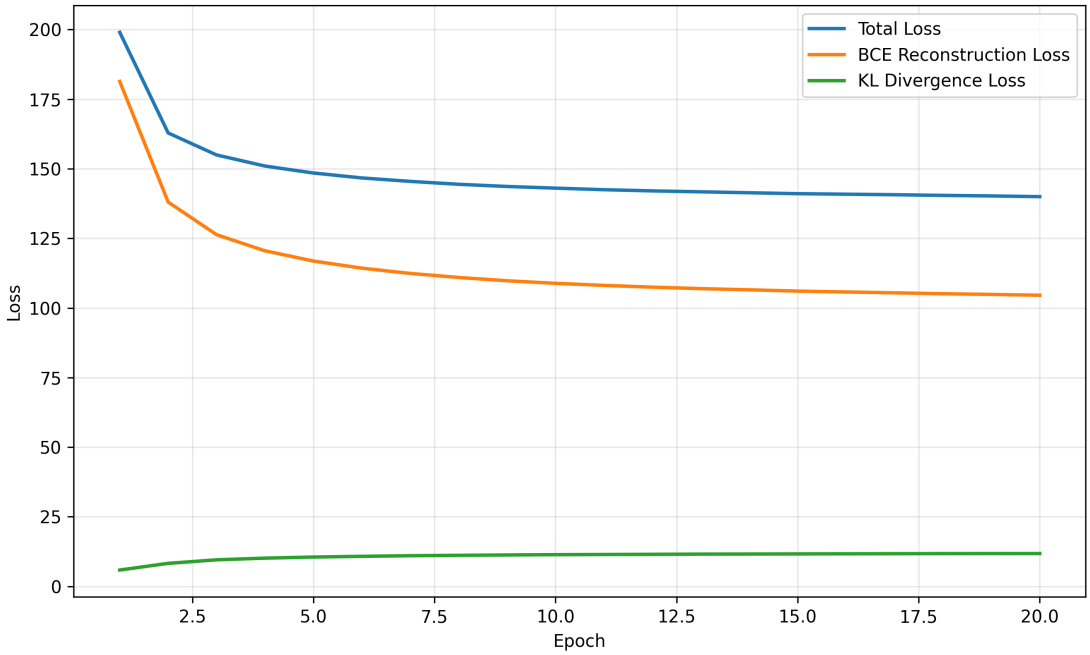


Figure 3.1: *Training loss for VAE with BCE and KLD.*

3.2 Generate New Samples

To conditionally generate an image from the MNIST dataset, I first searched the training dataset to find an image matching a target digit (7). Then, I sample an ϵ value matching the dimensionality of the latent space (20 by default) from a normal distribution. μ and σ samples are generated using the VAE encoder, and the latent vector is generated as

$$z = \mu + \sigma \odot \epsilon$$

as in the reparameterize method of the VAE class. The conditional images in Figure 3.3 are then generated using the decoder with the latent vector z as an input. With no conditional generation, as seen in Figure 3.2, it seems as if 4-5 ”8” digits were generated, showing limited diversity in the generation process. However, the clarity of those digits is not so good. Additional epochs did not improve training, which suggest some hyperparameter changes could benefit training. For conditional generation, all digits appeared as sevens, although some start to curve around to resemble a ”9”.

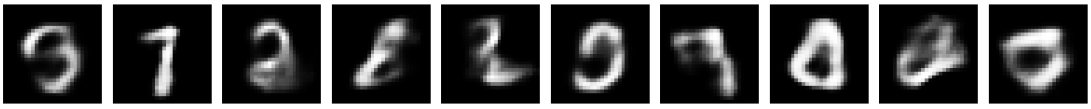


Figure 3.2: *10 generated images using the VAE trained with BCE loss.*

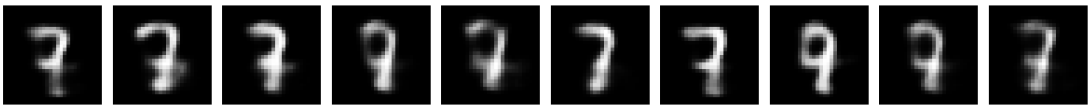


Figure 3.3: *10 images generated conditionally (target digit of 7) using the VAE trained with BCE loss.*

3.3 Modify the Loss Function

In this study, instead of using Binary Cross-Entropy Loss as the reconstruction loss function, Mean Square Error (MSE) loss is used instead. From the plot of the loss function over epochs, the magnitude of the loss is roughly 25% of the BCE loss. The quality of the generated digits is much better as well. Most of the unconditionally-generated images are distinguishable, though there are 4-5 ”9”s present. The quality of the conditionally-generated 7s are quite good, with all being distinguishable and only a single ”7” showing the loop of a ”9” digit.

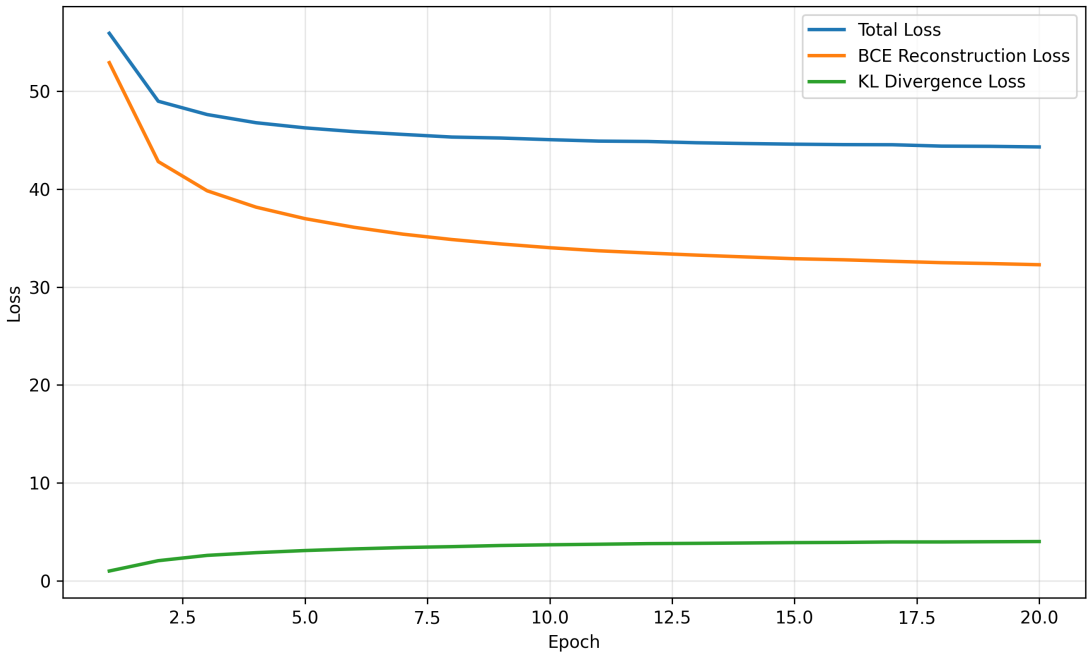


Figure 3.4: *Training loss for VAE with *MSE loss function.*

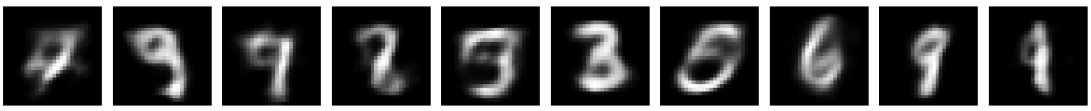


Figure 3.5: *10 generated images using the VAE trained with MSE loss.*

Figure 3.6: *10 images generated conditionally (target digit of 7) using the VAE trained with MSE loss.*

3.4 Latent Space Dimension

In this study, the latent space dimensionality used during training and generation is swept across values of [10,20,50], with 20 being the default dimension. In general, the larger the latent dimension, the slower the model converged over time (and the higher the loss), but all latent dimensions converged to a similar loss around epoch 20.

The quality of the conditionally-generated "7" digits was quite good for a latent space dimension of 10, with almost no artifacting of the "9" digit. The conditionally-generated "7" digits for the latent space of 50 are also quite good, and they even exhibit the slash through the middle of the "7", suggesting the larger latent space identified distinct features that other latent spaces did not.

For unconditionally-generated images, the model with a latent dimension of 50 seems to have 3-4 "8"s and 2-3 "9"s, and the model trained with a latent dimension of 10 also has a large number of "8"s. These similarities make sense, as the pattern of the loss function and ultimate final value are quite similar.

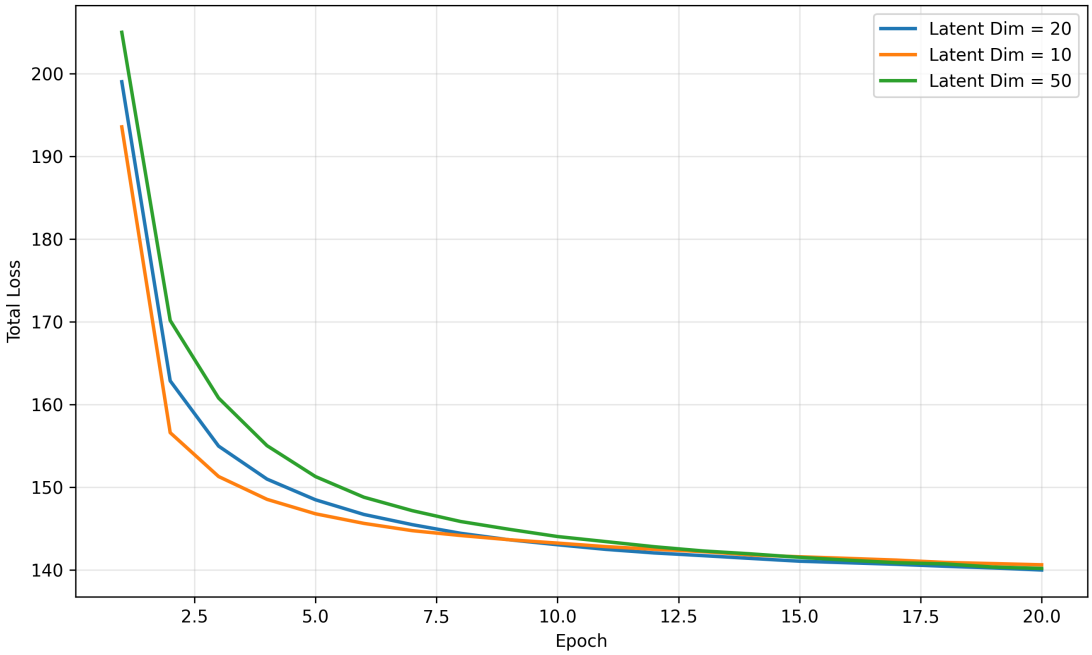


Figure 3.7: *Training loss for VAE with BCE loss and across latent space dimensions.*

Figure 3.8: *10 generated images using the VAE trained with BCE loss and latent space dimension of 10.*

Figure 3.9: *10 images generated conditionally (target digit of 7) using the VAE trained with BCE loss and latent space dimension of 10.*

Figure 3.10: *10 generated images using the VAE trained with BCE loss and latent space dimension of 50.*

Figure 3.11: *10 images generated conditionally (target digit of 7) using the VAE trained with BCE loss and latent space dimension of 50.*